



Published in final edited form as:

*Acta Biomater.* 2025 January 15; 192: 473–486. doi:10.1016/j.actbio.2024.11.040.

## Obesity Prolongs the Pro-Inflammatory Response and Attenuates Bone Healing on Titanium Implants

Derek Avery<sup>1</sup>, Lais Morandini<sup>1</sup>, Luke Sheakley<sup>1</sup>, Asmaa Alajmi<sup>1</sup>, Leah Bergey<sup>1</sup>, Henry J. Donahue<sup>1</sup>, Rebecca K. Martin<sup>2</sup>, Rene Olivares-Navarrete<sup>1,\*</sup>

<sup>1</sup>Department of Biomedical Engineering, College of Engineering, Virginia Commonwealth University, Richmond, VA, United States;

<sup>2</sup>Department of Microbiology and Immunology, School of Medicine, Virginia Commonwealth University, Richmond, VA, United States

### Abstract

Obesity is a metabolic disease resulting from excess body fat accumulation associated with chronic systemic inflammation. Obesity has been shown to impact the function and activity of neutrophils, macrophages, and T cells, contributing to higher circulating levels of pro-inflammatory cytokines. Biomaterial surface properties such as roughness and hydrophilicity can influence the behavior of immune cells in the peri-implant microenvironment. This study aimed to determine how obesity induced by a high-fat diet (HFD) affects the inflammatory response to modified titanium (Ti) implants and subsequent bone formation. Obese mice had significantly more neutrophils, pro-inflammatory macrophages, and T cells and fewer anti-inflammatory macrophages and mesenchymal stem cells (MSCs) in the peri-implant tissue than lean mice. Obesity also increased circulating adipokines and pro-inflammatory cytokines when compared to lean animals. Bone formation around Ti implants was reduced in obese mice compared to controls. Adoptive transfer of bone marrow cells isolated from obese mice into wild-type mice demonstrated the localized impact of obesity on immune cell function and phenotype, promoting a pro-inflammatory peri-implant microenvironment and attenuating bone formation post-implantation. These results show that obesity significantly affects the inflammatory response to modified Ti implants, prolonging the pro-inflammatory response to the implanted biomaterial and compromising bone formation.

### Graphical Abstract

\* **Corresponding Author:** Rene Olivares-Navarrete, DDS, PhD, VCU College of Engineering, Department of Biomedical Engineering, 70 S. Madison Street, Room 3328, Richmond, VA 23220, ronavarrete@vcu.edu.

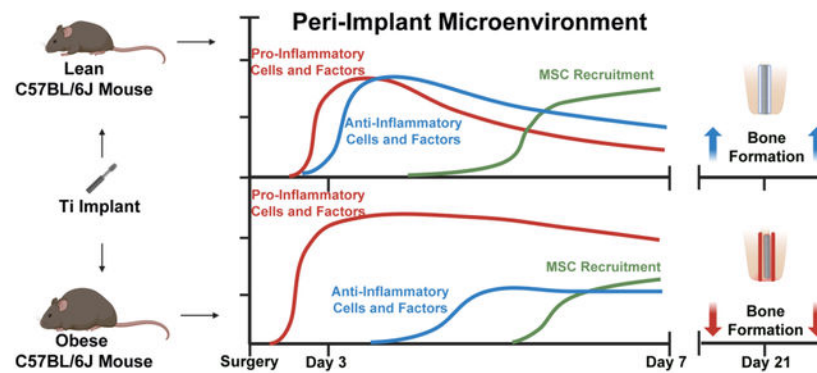
**Publisher's Disclaimer:** This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

Declaration of Competing Interest

The authors declare that they do not have competing financial interests.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.



## Keywords

Obesity; high-fat diet; inflammation; wound healing; titanium; biomaterials; bone formation

## 1. Introduction

Obesity is a metabolic disease that is the result of excessive body fat accumulation, negatively impacting health by increasing the risk of cardiovascular disease, type 2 diabetes, and osteoarthritis [1,2]. Obesity is a global epidemic with an estimated prevalence of 40% among the world population, and it has been linked to the development of chronic inflammation [2]. Likewise, pro-inflammatory cytokine production is increased in animal models of high-fat diet (HFD)-induced obesity [3,4]. Chronic inflammation due to obesity plays a vital role in altering the function of adipose tissue, contributing to systemic insulin resistance and increasing immune cell infiltration of adipose tissue [1,5,6]. Neutrophils, macrophages, and T cells are among the immune cells previously shown to infiltrate the adipose tissue and promote systemic inflammation [6–10]. Macrophages within the adipose tissue in HFD-induced obesity models have been suggested to be the primary source of inflammatory factors in obesity, demonstrating increased polarization towards a pro-inflammatory phenotype [11–13]. Diet-induced obesity impairs T cell response in mice, with higher numbers of Th17 cells and reduced Treg cells in lymph nodes and spleen compared to a standard, normal-fat diet [14–16].

Metallic implants used for craniofacial and orthopedic applications are modified at the surface level to promote bone formation, also known as osseointegration [17–19]. The success of biomaterial implantation depends upon the recruitment of immune cells and the resolution of the inflammatory response in the peri-implant microenvironment that results in biomaterial-tissue integration [20–23]. Post-implantation, damage to the tissue causes the release of damage-associated molecular patterns (DAMPs), initiating wound healing by recruiting immune cells to the implantation site. Innate immune cells, such as neutrophils and macrophages, are among the first immune cells recruited to the injury site, followed by T cells, as part of the adaptive response.

Neutrophils and macrophages are myeloid-derived cells that clear wound debris via phagocytosis and initiate the immune response [24,25]. Neutrophils release inflammatory

mediators like myeloperoxidase (MPO), neutrophil elastase, and neutrophil extracellular traps (NETs). Macrophages differentiate into pro- or anti-inflammatory macrophages depending on the biochemical and physical cues in the inflammatory microenvironment [25,26]. Our group has demonstrated that macrophage polarization is required to facilitate the amplification and resolution of the inflammatory response to implanted metallic biomaterials and orchestrate the recruitment of T cells and mesenchymal stem cells (MSCs) using titanium (Ti) as a model [22,25]. Ablating macrophages during the inflammatory response to Ti implants significantly attenuates T cell and MSC recruitment, compromising tissue regeneration [22].

T cells are immune cells found in lymphoid organs, recruited to the implantation site by chemokines and cytokines produced by neutrophils and macrophages at the biomaterial-tissue interface [21,23,25,27]. We have shown that T cells are polarized at the implantation site via direct interaction with macrophages and cytokines in the peri-implant environment [23,25,27]. CD4<sup>+</sup> T cells or T helper cells play an essential role in amplifying the immune response through interaction with antigen-presenting cells (*i.e.*, macrophages and dendritic cells) and by controlling the inflammatory microenvironment [27]. CD4<sup>+</sup> T cells can polarize into specific subsets, including Th1, Th2, Th17, and regulatory T (Treg) cells, dependent upon circulating cytokines in the inflammatory microenvironment [21,27,28]. We have previously shown that T cells are required to resolve the inflammatory response, recruit MSCs, and promote bone formation in response to biomaterial implantation [21,27]. While our prior findings were observed in animals of average body weight and composition, the significance of obesity in the immune response and bone formation remains understudied. Earlier studies on obesity and the foreign body response to implantation have found that obesity can attenuate the healing response, resulting in poor osseointegration [29,30].

This study aimed to elucidate the role of obesity in the immune cell response to biomaterial implantation and its impact on bone formation. We hypothesized that obesity promotes a pro-inflammatory microenvironment that increases the inflammatory response of neutrophils, macrophages, and T cells, reducing bone formation.

## 2. Materials and Methods

### 2.1 Ti Disks, Implants, and Material Characterization

Rough and rough-hydrophilic titanium (Ti) disks and implants were provided by Institut Straumann AG (Basel, Switzerland), as described previously [21,31,32]. 15 mm disks were fabricated from 1 mm thick unalloyed titanium grade 2. Disks were degreased in an acetone bath and treated in a 2% ammonium fluoride, 2% hydrofluoric acid, and 10% nitric acid solution to create the smooth Ti pretreatment. Rough Ti disks were fabricated by sand-blasting the pretreated smooth Ti with 0.25–0.50 mm coarse-grit corundum and acid etching in a HCl and H<sub>2</sub>SO<sub>4</sub> solution. After sand-blasting and acid etching, rough Ti disks were rinsed with deionized water and dried. Rough-hydrophilic Ti disks were produced following the same protocol to fabricate rough Ti disks but, to prevent air exposure, rinsed under a nitrogen atmosphere and sealed in an isotonic NaCl solution. 4 mm Ti implants were produced from 1 mm diameter rods and modified using the same procedure as disks. All materials were sterilized by  $\gamma$ -irradiation before use.

Scanning electron microscopy was used to qualitatively assess surface roughness (SEM, Zeiss Auriga, Carl Zeiss, Germany), and confocal microscopy was used for quantitative assessment (LSM 980 Laser Scanning Microscope, Germany) with a 20x objective and a total measurement area of  $790\text{ }\mu\text{m} \times 790\text{ }\mu\text{m}$  ( $n=3$ ). A moving average Gaussian filter with a  $30\text{ }\mu\text{m}$  cut-off wavelength was used to calculate the arithmetic mean height of the scale-limited surface ( $S_a$ ). Three samples were used for each surface type to calculate the mean surface roughness at six different areas. A Ramé-Hart goniometer was used to measure the sessile drop contact angle to determine surface hydrophilicity (Model 100–25a, Rame-Hart Instrument Co., Succasunna, NJ). 5  $\mu\text{L}$  drops of deionized water were used to measure surface hydrophilicity at six locations per sample ( $n=3$ ). X-ray photoelectron spectroscopy (XPS) was performed using a Phi5000 VersaProbe spectrometer (ULVAC-PHI, Inc.) to determine the oxide layer composition. A focused scanning monochromatic Al-K $\alpha$  source (1,486.6 eV) with a spot size of  $200\text{ }\mu\text{m}$  was used to acquire spectra, utilizing a base pressure of  $1 \times 10^{-7}$  Pa. FAT analyzer mode was used to run the instrument. Detail spectra of 187.85 eV and 46.95 eV were used for the pass energy for survey scans. CasaXPS was utilized for data analysis. Shirley background subtraction was used to integrate the signals. Measurements were taken on each sample in three separate areas.

## 2.2 HFD-induced Obesity Murine Model

20-week-old male C57BL/6J mice (The Jackson Laboratory, Stock #000664) were fed a diet containing 45 kcal% fat (D12451, Research Diets Inc., New Brunswick, NJ) or a standard diet containing 17 kcal% fat (Teklad LM-485, Inotiv, Madison, WI) for 6 weeks. Mice fed a 45 kcal% fat diet will be referred to as obese, and mice fed the standard diet will be referred to as lean. 26-week-old obese and lean mice were used for our experiments. The average weight of 26-week-old obese mice was 38.2 grams (standard deviation of 3.37 grams) compared to 22.6 grams (standard deviation of 1.66 grams) for lean mice. In this diet-induced obesity model, mice become pre-diabetic (Type 2) after eight weeks and diabetic after 16 weeks of a 45 kcal% fat diet [33]

## 2.3 Femoral Implantation Procedure

Isoflurane in O<sub>2</sub> gas was administered to mice to induce anesthesia, and mice were monitored for unconsciousness by pedal reflex. 1mg/kg buprenorphine ER-LAB (ZooPharm, Laramie, WY) was administered to mice before the surgical procedure to relieve postoperative pain for 72 hours. A scalpel was utilized to bisect the patellar tendon longitudinally and open the skin, overlaying the knee, until the femoral condyles were revealed. The medullary canal was accessed using a surgical dental bur, and a cylindrical rough or rough-hydrophilic Ti implant ( $\varnothing=1\text{mm}$ , 4mm length) was then press fit ( $n=6$  mice per implant condition). A digital X-ray system (NOMAD Pro 2, Dexis, Quakertown, PA) was used to confirm implant placement. Animals were monitored until initial ambulation and continually monitored every 24 hours. All animals were provided food and water *ad libitum* for the duration of the study. No signs of infection were present in this study. Animal procedures were used following a protocol approved by the Virginia Commonwealth University Institutional Animal Care and Use Committee (Protocol: AD10001108).

## 2.4 Impact of Obesity on Immune Response to Femoral Implants *In Vivo*

To determine the effect of obesity on the immune response to femoral implants, obese or lean mice were instrumented with femoral implants (n=6 mice per condition). Mice were euthanized on postoperative days 3 and 7 for cytokine and flow cytometry analysis and on day 21 for micro-computed tomography (microCT). Blood was collected via cardiac puncture and stored in a 1.5 mL tube. Blood was centrifuged at 500 g for 10 minutes, and plasma was isolated and stored at -80°C for analysis. Femurs were collected by blunt dissection. For peri-implant cell analysis, femurs were cut mid-shaft, and the implant and surrounding marrow were flushed using Accutase (Innovative Cell Technologies, San Diego, CA). For microCT analysis, femurs were fixed in a 10% formalin solution (ThermoFisher Scientific, Waltham, MA).

## 2.5 Flow Cytometry

Flow cytometry was conducted from bone marrow single-cell suspensions. ACK lysing buffer (ThermoFisher Scientific) was used to lyse red blood cells prior to incubation with antibodies. TruStain FcX (anti-CD16/32, BioLegend, San Diego, CA) was used to block Fc receptors. Antibodies specific for surface markers were used to identify neutrophils, total macrophages, pro-inflammatory macrophages, anti-inflammatory macrophages, MSCs, CD4<sup>+</sup> T cells, CD8<sup>+</sup> T cells, Th1, Th2, Th17, and Tregs (BioLegend, Supplemental Table 1). The gating strategy was established using single-color controls and fluorescent minus one, as previously described [27]. The samples were analyzed using a Cytex Aurora CS Cytometer (Cytex Biosciences, Fremont, CA). A minimum of 200,000 events were collected per sample, and FlowJo software (FlowJo LLC, Ashland, OR) was used for sample analysis.

## 2.6 Analysis of Adipokines and Cytokines in Serum

Levels of adiponectin, C-reactive protein (CRP), leptin, and resistin were measured in collected plasma by enzyme-linked immunosorbent assay (ELISA, RayBio, Peachtree Corners, GA) following the manufacturer's instructions. Circulating IL-1 $\beta$ , IL-4, IL-6, and TNF- $\alpha$  levels were measured by ELISA (BioLegend) following the manufacturer's instructions.

## 2.7 MicroCT Analysis

To understand how obesity impacts bone formation after biomaterial implantation, a resolution of 4032  $\times$  2688 pixels (voxel size of 7.00 $\mu$ m) was used to scan the distal femoral metaphysis over 180°. A 0.5mm high-resolution aluminum filter with an exposure time of 630 ms and 60kV and 200  $\mu$ A scanning energies was used. SkyScan 1276 (Bruker, Billerica, MA) was used every 0.15° to capture an image. SkyScan software and CTAn (Bruker) were used for the assembly and analysis of 3-D reconstructions from serial two-dimensional cross-sections. A tubular region of interest (ROI) was utilized in the distal portion of each femur to quantify de-novo bone formation. A volumetric sleeve aligned with the centroid of the implant was used to create the ROI, encompassing a length of 3.75 mm, an outer diameter of 1.12mm, and an inner diameter of 1.00mm. The ROI mask was applied to each image individually to evaluate all images. Bone was identified using a threshold of 1000.00–2500.00 Hounsfield units in the ROI. Dragonfly (Object Research Systems, Montréal, QC)

was used to create 3D representations of bone formation. Mineral density analysis was performed after calibration using bone calibration phantoms with different hydroxyapatite (HA) densities (0, 100, 200, 400, 800 mg HA/cm<sup>3</sup>) and density reported as tissue mineral density (TMD) in mg HA/cm<sup>3</sup>.

## 2.8 Macrophage Isolation

26-week-old C57BL/6J obese or lean mice were used to isolate and differentiate bone marrow-derived murine macrophages as described previously [21,27]. After flushing bone marrow from isolated femurs with PBS and treatment with ACK lysing buffer, RPMI 1640 (ThermoFisher Scientific) supplemented with 10% fetal bovine serum (ThermoFisher Scientific), 50 U/mL penicillin-50 µg/mL streptomycin (ThermoFisher Scientific), and 30 ng/mL macrophage colony-stimulating factor (M-CSF, BioLegend) was used to derive naïve macrophages. Cells were cultured at 37°C, 5% CO<sub>2</sub>, and 100% humidity. Fresh cell culture medium was supplemented on day 4 of culture. After 7 days, macrophages were detached with Accutase for use in subsequent experiments.

## 2.9 T Cell Isolation

26-week-old male C57BL/6J obese or lean mice were used to isolate and differentiate spleen-derived T cells, as previously described [21,27]. After harvest, spleens were homogenized, and ACK lysing buffer was used to remove red blood cells. Splenocytes were plated at a cell seeding density of 1,000,000 cells/well in 6-well plates to derive naïve T cells. Cells were cultured in RPMI 1640 Medium (ThermoFisher Scientific) supplemented with 10% fetal bovine serum (ThermoFisher Scientific), 50 U/mL penicillin-50 µg/mL streptomycin (ThermoFisher Scientific), 0.05 mM β-mercaptoethanol (ThermoFisher Scientific), 25 µL of prewashed and resuspended Dynabeads Mouse T-Activator CD3/CD28 (ThermoFisher Scientific), and 30 U/mL recombinant mouse IL-2 (BioLegend) for 3 days at 37°C, 5% CO<sub>2</sub>, and 100% humidity. Activated T cells were harvested after 3 days for use in subsequent experiments.

## 2.10 Effect of Obesity on Macrophage Response to Modified Ti *in Vitro*

To examine the effect of obesity on macrophage response to modified Ti, bone marrow-derived macrophages from obese mice were seeded in 24-well plates on rough or rough-hydrophilic Ti disks (n=6) at a density of 200,000 cells/well. Naïve macrophages isolated from lean mice were seeded on rough or rough-hydrophilic Ti substrates as controls. Macrophages were cultured for 24 hours before media was removed, and macrophages were detached with Accutase (Innovative Cell Technologies) for flow cytometry analysis. Flow cytometry was conducted and analyzed as described in Section 2.5 to identify pro-inflammatory and anti-inflammatory macrophages. A minimum of 50,000 events were collected for each sample.

## 2.11 Impact of Obesity on T Cell Polarization *In Vitro*

To understand the impact of obesity on T cell polarization in response to macrophages cultured to modified Ti *in vitro*, bone marrow-derived macrophages isolated from either obese or lean mice were seeded in 24-well plates on either rough or rough-hydrophilic



Ti disks (n=6) at a density of 200,000 cells/well. After 24 hours, activated T cells from obese-derived or lean-derived were added to the macrophage cultures at a seeding density of 300,000 cells/well. All cells were cultured in RPMI 1640 Medium (ThermoFisher Scientific) supplemented with 10% fetal bovine serum (ThermoFisher Scientific), 50 U/mL penicillin-50 µg/mL streptomycin (ThermoFisher Scientific), and 0.05 mM β-mercaptoethanol (ThermoFisher Scientific). Macrophages and T cells were co-cultured for 48 hours before T cells were isolated from the cell culture medium for flow cytometry analysis. T helper cell (Th1, Th2, Th17, Treg) subsets were analyzed using the flow cytometry methodology described above in section 2.5 and previously[27]. A minimum of 100,000 events were collected per sample for analysis.

## 2.12 Alteration of the Immune Cell Response by Adoptive Transfer of Bone Marrow from Obese Mice to Wild-Type Mice

To elucidate if the increased inflammatory phenotype observed in immune cells from obese mice persists in a normal environment, we performed an adoptive transfer of bone marrow cells from obese or lean mice into wild-type mice following Ti biomaterial implantation. Cells for adoptive transfer were isolated from the bone marrow of 26-week-old male C57BL/6J obese or lean mice. Mice were euthanized by CO<sub>2</sub> asphyxiation, and femur bones were removed. The bone marrow was flushed with PBS and treated with ACK buffer to remove erythrocytes (ThermoFisher Scientific). Cells were pooled and labeled with CellTracker™ Green CMFDA Dye (ThermoFisher Scientific) for adoptive transfer experiments.

26-week-old C57BL/6J lean mice were instrumented with rough Ti implants (n=6/condition) as described above. Immediately following the surgical procedure,  $1.5 \times 10^6$  fluorescently-tagged bone marrow cells were injected in 100 µL of PBS via the tail vein. Mice were fed a standard diet *ad libitum* following surgery. Mice were euthanized on postoperative days 3 and 7 for cytokine/adipokine and flow cytometry analysis and on day 21 for microCT.

## 2.13 Statistical Analysis

Data are presented as mean ± SD. Prism10 (GraphPad Software, San Diego, CA) was used for statistical analysis. The Shapiro-Wilk normality test was first performed on data to determine if data was normally distributed. The null hypothesis that group means were equal at a significance level of  $\alpha=0.05$  was tested using one-way or two-way analysis of variance (ANOVA). One-way ANOVA was utilized for baseline assessment of immune cells, adipokines, and cytokines between lean and obese mice prior to implantation. One-way ANOVA was also performed to compare the effect of adoptively transferring obese-derived or lean-derived bone marrow to analyze secretion of adipokines and cytokines, bone implant contact, and tissue mineral density. Two-way ANOVA was performed for *in vitro* and *in vivo* experiments involving rough or rough-hydrophilic Ti to compare the effect of both obesity and surface hydrophilicity. Multiple comparisons were conducted using post-hoc Tukey's HSD test.

### 3. Results

#### 3.1 Titanium Disks and Implant Characterization

Rough and rough-hydrophilic Ti surfaces were similar in surface roughness, both quantitatively and qualitatively (Figure 1A). Rough-hydrophilic Ti ( $\theta = 0^\circ$ ) was more hydrophilic, and rough Ti ( $\theta = 93^\circ$ ) was more hydrophobic, as determined by analysis of the water contact angle (Figure 1B). The wettability of rough and rough-hydrophilic Ti surfaces was correlated with the quantity of carbon in the oxide layer. Rough Ti demonstrated a higher quantity of carbon in the oxide layer than rough-hydrophilic Ti.

#### 3.2 Obesity Promotes Pro-Inflammatory Innate Immune Response within Peri-implant Microenvironment *In Vivo*

At baseline prior to implantation, obese mice demonstrated higher proportions of neutrophils and pro-inflammatory macrophages compared to lean mice (Supplemental Figure 2). Total macrophages and anti-inflammatory macrophages were similar between lean and obese mice.

On day 3 post-implantation, lean mice receiving rough Ti implants showed higher populations of neutrophils, total macrophages, and pro-inflammatory macrophages than rough-hydrophilic Ti (Figure 2). Lean mice receiving rough-hydrophilic Ti implants showed higher populations of anti-inflammatory macrophages when compared to rough Ti. Obese mice demonstrate an altered innate immune response to rough and rough-hydrophilic Ti. On postoperative day 3, total macrophages and pro-inflammatory macrophages were significantly higher in obese mice than in lean mice, while anti-inflammatory macrophages in obese mice were lower.

At 7 days post-implantation, lean mice receiving rough Ti implants demonstrated higher populations of neutrophils, total macrophages, and pro-inflammatory macrophages than mice implanted with rough-hydrophilic Ti (Figure 2). Obese mice had higher populations of neutrophils, total macrophages, and pro-inflammatory macrophages than lean mice. Flow cytometry scatter plots for neutrophils, total macrophages, and pro- and anti-inflammatory macrophages at day 7 post-implantation can be found in Supplementary Figure 4.

#### 3.3 Obesity Alters Recruitment of T Cells and MSCs in the Peri-implant Microenvironment *In Vivo*

Next, we analyzed the effect of obesity on the recruitment of T cells and MSCs in response to modified Ti implantation. Prior to implantation, obese mice showed higher proportions of CD8<sup>+</sup> T cells and lower CD4<sup>+</sup> T cells when compared to lean mice (Supplemental Figure 2). At day 3 post-implantation, we did not note differences in populations of CD4<sup>+</sup> T cells, CD8<sup>+</sup> T cells, and MSCs between implant surfaces (Figure 3). When compared to lean mice, obese mice demonstrated lower populations of CD4<sup>+</sup> T cells and MSCs. Conversely, CD8<sup>+</sup> T cells were significantly higher in obese mice compared to lean mice.

On postoperative day 7, mice receiving rough-hydrophilic Ti implants showed lower populations of CD8<sup>+</sup> T cells and higher recruitment of MSCs (Figure 3). CD4<sup>+</sup> T cells did



not demonstrate differences between rough and rough-hydrophilic Ti implants. Obese mice demonstrated higher populations of CD4<sup>+</sup> and CD8<sup>+</sup> T cells when compared to lean mice. MSCs in obese mice were significantly lower than in lean mice. Flow cytometry scatter plots for T cells and MSCs at post-operative day 7 can be found in Supplementary Figure 5.

### **3.4 Obesity Modulates Helper T Cell Polarization in the Peri-implant Microenvironment *In Vivo***

We also were interested in how obesity affects the polarization of CD4<sup>+</sup> T cell subsets that contribute to inflammation in the peri-implant microenvironment. Pre-implantation, obese mice showed higher polarization of Th1 and CD4<sup>+</sup> Tregs than lean mice (Supplemental Figure 3). Populations of Th2 and Th17 were significantly lower in obese mice than in lean mice.

At 3 days post-implantation, mice receiving rough-hydrophilic Ti implants showed lower Th1 and Th17 cells and higher populations of Th2 and CD4<sup>+</sup> Tregs (Figure 4). Obese mice demonstrated significantly higher polarization of Th1 and Th17 subsets when compared to lean mice. Additionally, obese mice polarized less CD4<sup>+</sup> T cells to Th2 and Tregs. Interestingly, the polarization of Th1, Th17, and Treg subsets from obese mice was similar between mice receiving rough and rough-hydrophilic Ti implants.

On day 7 post-implantation, lean mice receiving rough-hydrophilic Ti implants showed lower polarization of Th1, Th2, Th17, and CD4<sup>+</sup> Tregs than rough Ti (Figure 4). Obese mice demonstrated significantly higher polarization to Th1 and Th17 subsets and reduced polarization of Th2 and Tregs compared to lean mice. Obese mice receiving either modified Ti implant showed similar polarization of Tregs. Flow cytometry scatter plots for CD4<sup>+</sup> T cell subsets at day 7 post-implantation can be found in Supplementary Figure 6.

### **3.5 Systemic Adipokine and Cytokine Production is Significantly Skewed by Obesity *In Vivo***

Next, we examined the impact of obesity on the systemic production of adipokines and cytokines. At baseline, serum levels of leptin, resistin, and CRP were significantly higher in obese mice than in lean mice (Figure 5). Production of adiponectin was significantly lower at baseline in obese mice. Additionally, serum levels of pro-inflammatory cytokines IL-1 $\beta$ , IL-6, IL-17A, and TNF- $\alpha$  were significantly higher in obese mice.

On day 3 post-implantation, serum levels of adiponectin, CRP, and resistin in lean mice were similar between different modified Ti implants (Figure 6). Leptin was lower in lean mice receiving rough-hydrophilic implants compared to rough Ti. IL-1 $\beta$ , IL-6, IL-17A, and TNF- $\alpha$  levels were lower in mice receiving rough-hydrophilic Ti implants. Serum from obese mice had higher levels of CRP, leptin, resistin, IL-1 $\beta$ , IL-6, IL-17A, and TNF- $\alpha$  than that from lean mice (Figure 6). In obese mice, levels of these proteins were lower in mice receiving rough-hydrophilic Ti implants. Adiponectin was significantly lower in obese mice.

At day 7 post-implantation, CRP, leptin, and resistin did not show differences between lean mice receiving different implants (Figure 7). Mice receiving rough-hydrophilic Ti implants showed higher levels of adiponectin and lower levels of IL-1 $\beta$ , IL-6, IL-17A, and TNF- $\alpha$ . In

obese mice, CRP, leptin, resistin, IL-1 $\beta$ , IL-6, IL-17A, and TNF- $\alpha$  levels were significantly higher when compared to lean mice. Adiponectin was significantly lower in obese mice, although these levels did not differ between rough and rough-hydrophilic Ti implants.

### 3.6 Bone Formation is Significantly Attenuated by Obesity *In Vivo*

To understand if obesity affects bone formation, we analyzed bone formation at 21 days post-implantation (Figure 8). In lean mice, bone formation was higher on rough-hydrophilic Ti implants than on rough Ti. Obese mice produced significantly less bone around Ti implants than lean mice, and the effect of implant hydrophilicity was abrogated (Figure 8). Additionally, we analyzed tissue mineral density and found that tissue mineral density was higher in lean mice receiving rough-hydrophilic Ti implants when compared to rough Ti (Supplemental Figure 1A). Obese mice demonstrated significantly lower quantities of tissue mineral density when compared to lean mice.

### 3.7 Obesity Perturbs Helper T Cell Polarization in Macrophage-T Cell Co-cultures *In Vitro*

Next, we used a co-culture model to examine how obesity induces polarization of T cell subsets *in vitro*. We analyzed the impact of obese-derived macrophages on the polarization of lean-derived T cells and alterations in T cell polarization when obese-derived T cells were cultured with lean-derived macrophages. Polarization of Th1 and Th17 cells was lower in response to lean-derived macrophages cultured on rough-hydrophilic Ti surfaces in comparison to rough Ti (Figure 9). Th2 and Treg polarization was the highest in response to lean-derived macrophages cultured on rough-hydrophilic Ti when compared against rough Ti.

Analyzing the effect of obesity on T cell polarization, Th1 and Th17 cells were higher in response to obese-derived macrophages compared to co-culture with lean-derived macrophages (Figure 9). Treg populations were lower in response to obese-derived macrophages when compared to Tregs cultured with lean-derived macrophages.

Examining the polarization of obese-derived T cells cultured with lean-derived macrophages, we found that Th1 cells were lower than co-cultures with obese-derived macrophages (Figure 9). Th17 cell polarization was dependent on surface hydrophilicity, but obese-derived Th17 cells were higher than co-cultures with lean-derived macrophages and T cells. The polarization of obese-derived Th2 and Treg cells was significantly attenuated compared to other groups with lean-derived T cells.

### 3.8 Obesity Promotes Pro-Inflammatory Macrophage Phenotype *In Vitro*

To understand the effect of obesity on the polarization of macrophages *in vitro*, we cultured bone marrow-derived macrophages from obese or lean mice on rough and rough-hydrophilic Ti disks. The proportion of lean-derived pro-inflammatory macrophages was lower in response to rough-hydrophilic Ti surfaces than rough Ti (Figure 10). In contrast, anti-inflammatory macrophages were higher in response to rough-hydrophilic Ti. Obese-derived pro-inflammatory macrophages were significantly elevated in response to both surfaces, with the highest populations occurring in response to rough Ti. Obese-derived

anti-inflammatory macrophages were lower on all substrates when compared to lean-derived macrophages.

### 3.9 Adoptive Transfer of Bone Marrow from Obese Mice Impacts the Native Immune Cells in the Peri-Implant Microenvironment

To understand the cellular response of obesity on native immune cells in the inflammatory microenvironment of wild-type mice, we instrumented 26-week-old mice with rough Ti femoral implants and adoptively transferred bone marrow cells from obese or lean mice. We performed immunophenotyping analysis via flow cytometry on postoperative day 3 or 7. At day 3 post-implantation, populations of neutrophils, total macrophages, and pro-inflammatory macrophages did not demonstrate significant differences between mice receiving either obese or lean-derived bone marrow (Figure 11). Anti-inflammatory macrophages were lower in mice receiving obese-derived bone marrow. CD4<sup>+</sup> and CD8<sup>+</sup> T cells were significantly higher in response to obese-derived cells. MSCs were similar between mice receiving either obese or lean-derived bone marrow.

At day 7 post-implantation, neutrophils, total macrophages, and pro-inflammatory macrophages were significantly higher in mice receiving obese-derived bone marrow (Figure 11). Anti-inflammatory macrophages and MSCs were lower in mice receiving obese-derived bone marrow when compared to lean-derived cells. CD4<sup>+</sup> and CD8<sup>+</sup> T cells were higher in response to obese-derived bone marrow but did not demonstrate significant differences from 3 days post-implantation (Figure 12).

### 3.10 Adoptive Transfer of Bone Marrow from Obese Mice Significantly Alters Systemic Production of Pro-inflammatory Factors *In Vivo*

To elucidate the impact of changes in cell function induced by obesity on systemic protein secretion, we collected blood from mice that were adoptively transferred bone marrow cells from lean or obese mice after instrumentation with rough Ti implants. We analyzed plasma at 3- and 7-days post-implantation to quantify protein production. At day 3 post-implantation, CRP, IL-1 $\beta$ , IL-6, IL-17A, and TNF- $\alpha$  were significantly higher in mice receiving obese-derived cells than lean-derived cells (Figure 13). Serum levels of adiponectin, leptin, and resistin were similar between animals receiving obese or lean-derived bone marrow.

On postoperative day 7, serum levels of CRP, IL-1 $\beta$ , IL-6, IL-17A, and TNF- $\alpha$  remained higher in mice receiving obese-derived cells (Figure 14). Serum levels of adiponectin, leptin, and resistin did not demonstrate significant differences between mice receiving either obese or lean-derived cells.

### 3.11 Adoptive Transfer of Bone Marrow from Obese Mice Significantly Decreases Bone Formation Around Ti Implants

To elucidate the effect of adoptive transfer of immune cells from obese mice on bone formation in wild-type mice, we analyzed bone formation at 21 days post-implantation. Mice that were adoptively transferred obese-derived bone marrow demonstrated significantly less bone formation around Ti implants when compared to mice

receiving lean-derived cells (Figure 15). Examining tissue mineral density, we found that mice receiving obese-derived bone marrow also demonstrate reduced tissue mineral density when compared to adoptive transfer of lean-derived cells (Supplemental Figure 1B)

#### 4. Discussion

Biomaterial surface modifications like roughness and wettability can dictate the recruitment and behavior of immune cells at the implant-tissue interface. Previously, we and others have shown that innate immune cells, including neutrophils and macrophages, activate and polarize in response to differences in Ti physicochemical cues *in vitro* and *in vivo* [22,25,32,34,35]. Additionally, we have shown that T cell polarization *in vitro* and *in vivo* also responds to changes in biomaterial surface roughness and wettability through interactions with macrophages [21,25,27]. This study demonstrates that environmental factors such as obesity significantly impact the inflammatory response and polarization of immune cells within the peri-implant microenvironment. Our data shows that neutrophils, macrophages, T cells, and MSC recruitment are significantly affected by obesity. Here, we utilized rough and rough-hydrophilic Ti implants to demonstrate that obesity attenuates the anti-inflammatory and pro-osseointegrative response associated with increasing surface wettability on modified Ti implants.

The inflammatory response is initiated by DAMPs that chemoattract neutrophils to the injury site. While the response of neutrophils in infection and disease has been well characterized, the role of neutrophils in response to biomaterial implantation remains unclear. Neutrophils are activated in the onset of diet-induced obesity as early as three days of HFD feeding, producing significantly higher levels of MPO [5,36,37]. MPO is a heme peroxidase superfamily member that catalyzes the formation of numerous reactive oxidants and diffusible radical species [38]. Persistent activation of neutrophils can result in higher production of MPO, NETs, and other inflammatory mediators, damaging the host tissue and further promoting inflammation [39,40]. Higher activation of neutrophils in animal models of diet-induced obesity has also been shown to increase the infiltration of macrophages [5,37,39]. Our study demonstrates a higher proportion of neutrophils in obese mice, showing that these cells play a pivotal role in exacerbating the inflammatory response in the presence of obesity.

Macrophages play a significant role in the inflammatory response to implants and diet-induced obesity. Macrophages are the most abundant cell type that infiltrates the adipose tissue, and prior studies have shown that macrophages in animal models of diet-induced obesity have an abnormal oxidative burst and altered phagocytotic activity that can contribute to exacerbated inflammatory response [11,12]. Our study shows that obesity impacts the polarization of macrophages, promoting higher proportions of pro-inflammatory macrophages pre- and post-implantation, as well as lower polarization of anti-inflammatory macrophages. Additionally, macrophages in animal models of diet-induced obesity have shown higher production of cytokines, including TNF- $\alpha$ , IL-1 $\beta$ , and IL-6, as well as nitric oxide and monocyte chemoattractant protein-1 (MCP-1) when compared to mice that were of standard weight and body composition [41,42]. Our study showed significantly higher serum levels of pro-inflammatory cytokines, including CRP, in obese mice compared to lean

mice. CRP is strongly associated with obesity in human studies [43,44]. While CRP has been known as an inflammation marker, recent studies have shown that CRP can activate the innate immune system. Chronic elevation of CRP increases complement component levels and modulates blood counts of innate immune cells [44]. A previous *in vitro* model involving cell culture of monocytes with CRP-primed monocytes to differentiate into pro-inflammatory macrophages promoted the conversion of anti-inflammatory macrophages to pro-inflammatory macrophages and upregulated the production of TNF- $\alpha$  and IL-1 $\beta$  [45]. Our results show higher macrophage numbers in obese mice than lean mice, and obesity skewed macrophages towards a pro-inflammatory phenotype. Our findings also demonstrate that obesity elevates pro-inflammatory macrophages at all points analyzed, exhibiting an impaired transition to an anti-inflammatory phenotype observed in lean mice.

Like neutrophils and macrophages, T cells have also been investigated in animal models of HFD-induced obesity but have not been explored in the context of bone healing in obese subjects. Both CD4<sup>+</sup> and CD8<sup>+</sup> T cell subsets are essential for resolving inflammation and bone regeneration in the context of implantation, as the ablation of these subsets results in reduced MSC recruitment and proliferation and reduced bone formation around the biomaterial implant [21,27]. CD4<sup>+</sup> and CD8<sup>+</sup> T cell populations are significantly higher in models of HFD-induced obesity, and these subsets have demonstrated different roles in amplifying systemic inflammation [8–10,16,46]. CD8<sup>+</sup> T cell populations are present in higher quantities in mice fed HFD, infiltrating and interacting with adipose tissue before other immune cells [8,47]. While it is unknown why these cells are recruited in the adipose tissue before other cell types, CD8<sup>+</sup> T cells increase the recruitment of pro-inflammatory macrophages and the production of TNF- $\alpha$  [8,9]. Elevated CD8<sup>+</sup> T cell populations are associated with poor fracture healing and lack of bone formation [48]. In our study, CD8<sup>+</sup> T cells were higher in obese mice prior to implantation, as well as all time points post-implantation. Similarly, CD4<sup>+</sup> T cells were higher in obese mice on day 7 post-implantation but lower at earlier times. Prior studies examining CD4<sup>+</sup> T cells in obesity observe significantly lower production of IL-2, which impairs the recruitment and proliferation of CD4<sup>+</sup> T cells in the early inflammatory process [49,50].

Variance in the polarization of CD4<sup>+</sup> T cells provides more context to obesity-induced alterations of the inflammatory response. Th1 and Th17 cells have shown significant elevations in models of diet-induced obesity, along with increased production of their signature pro-inflammatory cytokines, IFN- $\gamma$ , and IL-17 [10,14,46]. We have demonstrated similar results in our study, showing significantly higher polarization of Th1 and Th17 cells and significantly higher production of the IL-17 isoform IL-17A in obese mice. Our findings show enhanced Th1/Th17 polarization at postoperative day 3, which remains elevated at day 7, demonstrating a prolonged pro-inflammatory T cell response in obese mice. Additionally, we showed that leptin, a hormone derived from adipose tissue, was significantly upregulated in obese mice, which has demonstrated a pro-inflammatory role in the immune system in other studies [51,52]. Leptin receptors are highly expressed on T cells, specifically Tregs, and leptin promotes T cell proliferation and increases pro-inflammatory cytokine secretion, skewing T cell polarization towards Th1/Th17 phenotypes and reducing polarization to Th2/Treg phenotypes [14,16,46]. Tregs are essential for the suppression of the inflammatory response, as prior work suggests that Tregs are more significantly impacted by diet-induced

obesity in comparison to other CD4<sup>+</sup> T cell subsets, producing higher levels of reactive oxygen species [16,46,53–55]. Our study demonstrated higher polarization of Tregs in obese mice prior to implantation, with indiscernible changes to the proportion of Tregs through the inflammatory time course, suggesting that the exacerbated inflammatory response associated with obesity may result from inadequate immunomodulation by Tregs.

MSC recruitment is needed as part of the regenerative process orchestrated by immune cells [21,22,25,27]. Prior obesity studies have noted decreased differentiation of MSCs to osteoblasts, resulting in reduced bone formation and enhanced fibrous encapsulation [56,57]. Additionally, mice fed a HFD have shown reduced populations of bone marrow MSCs, as well as increased differentiation to adipogenic lineages [56,58–61]. Adiponectin is an adipokine that is typically reduced in cases of type 2 diabetes and obesity, and a prior murine model of diet-induced obesity involving a calvarial bone defect found that adiponectin promotes the migration of bone marrow-derived MSCs, as well as bone formation [62,63]. Our study found that adiponectin is significantly reduced in obese mice, coinciding with attenuated MSC recruitment and bone formation. These findings align with a previous murine study that found that obesity significantly delayed bone healing after fracture [64]. We also note that obese mice exhibited lower tissue mineral density in new bone. Bone mineral density has been analyzed previously in obese mice, demonstrating decreased femoral bone volume and quality, suggesting increased bone resorption via heightened leptin or pro-inflammatory cytokine secretion in response to obesity [65–67].

While it is known that obesity produces chronic inflammation with an increased number of immune cells, a recent study showed that obesity produces persistent changes in innate immunity even after weight loss and normalization of metabolism [68]. The investigators found epigenetic changes in chromatin in immune cells associated with increased expression of genes that function in inflammatory responses [68]. To assess the effect of possible epigenetic alterations in immune cells exposed to HFD-induced obesity in the resolution of the inflammatory response and biomaterial integration, we performed experiments using the adoptive transfer of bone marrow immune cells into lean mice instrumented with rough Ti implants. Our results showed clear differences in the resolution of the inflammatory response where mice treated with cells isolated from obese mice failed to resolve the inflammatory response compared to mice receiving cells isolated from lean mice. Our data suggests that immune cells from obese mice acquire an altered function or phenotype contributing to cell behavior in the peri-implant microenvironment, even when adoptively transferred into lean mice. These alterations in immune cell function may result from epigenetic changes induced through obesity that impact cells at the transcriptional level. Prior work in assessing the role of nutrition in epigenetic modifications has shown that dietary changes can alter gene expression via inhibition of enzymes specific to modifications of histones or methylation of DNA, ultimately altering the phenotype of the cell [69–71]. Epigenetic changes have been identified in immune cells in previous obesity studies, specifically in T cells and macrophages [68,70,72–74]. Macrophages can be primed to differentiate towards a pro-inflammatory phenotype because of these changes, previously demonstrated in a prior murine model of diet-induced obesity, showing that these alterations occur at the bone marrow progenitor cell level [68,70,73,74]. T cells are highly susceptible to epigenetic changes because of dietary changes, as changes in methylation



in the transcription factors that distinguish CD4<sup>+</sup> T cell subsets can dictate the degree of polarization towards these subsets [70,72,73]. We show that obesity promotes immune cell dysfunction in the peri-implant microenvironment, resulting in higher secretion of pro-inflammatory factors and reduced or delayed healing at the implantation site.

## 5. Conclusion

Our results show that the inflammatory response to Ti implants is significantly skewed in HFD-induced obesity, with a higher presence of pro-inflammatory factors and immune cells contributing to a chronic inflammatory peri-implant microenvironment. Additionally, macrophage and T cell polarization is shifted towards pro-inflammatory phenotypes, further influencing the inflammatory microenvironment and reducing MSC recruitment and bone formation. Furthermore, our adoptive transfer experiments demonstrated a localized effect of cells isolated from obese mice in the peri-implant microenvironment, suggesting an alteration in the immune cell phenotype and function after the development of obesity via HFD. These findings highlight the impact of obesity on inflammation and healing, dictating the degree of bone formation around the biomaterial implant and, ultimately, the rate of success of implants.

## Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

## Acknowledgment

Research reported in this publication was supported by the National Institute of Dental and Craniofacial Research of the National Institutes of Health under award number R01DE028919. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health. Institute Straumann AG (Basel, Switzerland) supplied Ti implants and surfaces. Services in support of the research project were provided by the VCU Massey Cancer Center Flow Cytometry Shared Resource supported, in part, with funding from NIH-NCI Cancer Center Support Grant P30 CA016059. The raw/processed data required to reproduce these findings cannot be shared at this time as it is also part of an ongoing study.

## References

- [1]. Xu H, Barnes GT, Yang Q, Tan G, Yang D, Chou CJ, Sole J, Nichols A, Ross JS, Tartaglia LA, Chen H, Chronic inflammation in fat plays a crucial role in the development of obesity-related insulin resistance, *J Clin Invest* 112 (2003) 1821–1830. 10.1172/JCI19451. [PubMed: 14679177]
- [2]. Jaacks LM, Vandevijvere S, Pan A, McGowan CJ, Wallace C, Imamura F, Mozaffarian D, Swinburn B, Ezzati M, The obesity transition: stages of the global epidemic, *Lancet Diabetes Endocrinol* 7 (2019) 231–240. 10.1016/S2213-8587(19)30026-9. [PubMed: 30704950]
- [3]. Baumgarner KM, Setti S, Diaz C, Littlefield A, Jones A, Kohman RA, Diet-induced obesity attenuates cytokine production following an immune challenge, *Behavioural Brain Research* 267 (2014) 33–41. 10.1016/J.BBR.2014.03.017. [PubMed: 24657736]
- [4]. Mito N, Hosoda T, Kato C, Sato K, Change of cytokine balance in diet-induced obese mice, *Metabolism* 49 (2000) 1295–1300. 10.1053/META.2000.9523. [PubMed: 11079819]
- [5]. Nijhuis J, Rensen SS, Slaats Y, Van Dielen FMH, Buurman WA, Greve JWM, Neutrophil Activation in Morbid Obesity, *Chronic Activation of Acute Inflammation, Obesity* 17 (2009) 2014–2018. 10.1038/OBY.2009.113. [PubMed: 19390527]
- [6]. Liu Y, Lu X, Li X, Du P, Qin G, High-fat diet triggers obesity-related early infiltration of macrophages into adipose tissue and transient reduction of blood monocyte count, *Mol Immunol* 117 (2020) 139–146. 10.1016/J.MOLIMM.2019.11.002. [PubMed: 31778913]



- [7]. Ou R, Liu J, Lv M, Wang J, Wang J, Zhu L, Zhao L, Xu Y, Neutrophil depletion improves diet-induced non-alcoholic fatty liver disease in mice, *Endocrine* 57 (2017) 72–82. 10.1007/S12020-017-1323-4/FIGURES/5. [PubMed: 28508193]
- [8]. Nishimura S, Manabe I, Nagasaki M, Eto K, Yamashita H, Ohsugi M, Otsu M, Hara K, Ueki K, Sugiura S, Yoshimura K, Kadowaki T, Nagai R, CD8<sup>+</sup> effector T cells contribute to macrophage recruitment and adipose tissue inflammation in obesity, *Nature Medicine* 2009 15:8 15 (2009) 914–920. 10.1038/nm.1964.
- [9]. Rausch ME, Weisberg S, Vardhana P, Tortoriello DV, Obesity in C57BL/6J mice is characterized by adipose tissue hypoxia and cytotoxic T-cell infiltration, *International Journal of Obesity* 2008 32:3 32 (2007) 451–463. 10.1038/sj.ijo.0803744.
- [10]. Strissel KJ, DeFuria J, Shaul ME, Bennett G, Greenberg AS, Obin MS, T-Cell Recruitment and Th1 Polarization in Adipose Tissue During Diet-Induced Obesity in C57BL/6 Mice, *Obesity* 18 (2010) 1918–1925. 10.1038/OBY.2010.1. [PubMed: 20111012]
- [11]. Liu K, Zhao E, Ilyas G, Lalazar G, Lin Y, Haseeb M, Tanaka KE, Czaja MJ, Impaired macrophage autophagy increases the immune response in obese mice by promoting proinflammatory macrophage polarization, *Autophagy* 11 (2015) 271–284. 10.1080/15548627.2015.1009787. [PubMed: 25650776]
- [12]. Zhou Q, Leeman SE, Amar S, Signaling mechanisms involved in altered function of macrophages from diet-induced obese mice affect immune responses, *Proc Natl Acad Sci U S A* 106 (2009) 10740–10745. 10.1073/PNAS.0904412106. [PubMed: 19541650]
- [13]. Lumeng CN, DeYoung SM, Bodzin JL, Saltiel AR, Increased Inflammatory Properties of Adipose Tissue Macrophages Recruited During Diet-Induced Obesity, *Diabetes* 56 (2007) 16–23. 10.2337/DB06-1076. [PubMed: 17192460]
- [14]. Winer S, Paltser G, Chan Y, Tsui H, Engleman E, Winer D, Dosch HM, Obesity predisposes to Th17 bias, *Eur J Immunol* 39 (2009) 2629–2635. 10.1002/EJI.200838893. [PubMed: 19662632]
- [15]. Magnuson AM, | Regan DP, | Fouts JK, | Booth AD, | Dow SW, | Foster MT, Foster MT, Diet-induced obesity causes visceral, but not subcutaneous, lymph node hyperplasia via increases in specific immune cell populations, *Cell Prolif* (2017) 50. 10.1111/cpr.12365.
- [16]. Zhao Y, Lin L, Li J, Xiao Z, Chen B, Wan L, Li M, Wu X, Hin Cho C, Shen J, CD4<sup>+</sup> T cells in obesity and obesity-associated diseases, *Cell Immunol* 332 (2018) 1–6. 10.1016/J.CELLIMM.2018.08.013. [PubMed: 30146083]
- [17]. Agarwal R, García AJ, Biomaterial strategies for engineering implants for enhanced osseointegration and bone repair, *Adv Drug Deliv Rev* 94 (2015) 53–62. 10.1016/J.ADDR.2015.03.013. [PubMed: 25861724]
- [18]. Boyan BD, Cheng A, Olivares-Navarrete R, Schwartz Z, Implant Surface Design Regulates Mesenchymal Stem Cell Differentiation and Maturation, *Adv Dent Res* 28 (2016) 10–17. 10.1177/0022034515624444. [PubMed: 26927483]
- [19]. Olivares-Navarrete R, Hyzy SL, Hutton DL, Erdman CP, Wieland M, Boyan BD, Schwartz Z, Direct and indirect effects of microstructured titanium substrates on the induction of mesenchymal stem cell differentiation towards the osteoblast lineage, *Biomaterials* 31 (2010) 2728–2735. 10.1016/J.BIOMATERIALS.2009.12.029. [PubMed: 20053436]
- [20]. Selders GS, Fetzi AE, Radic MZ, Bowlin GL, An overview of the role of neutrophils in innate immunity, inflammation and host-biomaterial integration, *Regen Biomater* 4 (2017) 55–68. 10.1093/RB/RBW041. [PubMed: 28149530]
- [21]. Avery D, Morandini L, Sheakley L, Grabiec M, Olivares-Navarrete R, CD4<sup>+</sup> and CD8<sup>+</sup> T cells reduce inflammation and promote bone healing in response to titanium implants, *Acta Biomater* 179 (2024) 385–397. 10.1016/J.ACTBIO.2024.03.022. [PubMed: 38554889]
- [22]. Abaricia JO, Shah AH, Chaubal M, Hotchkiss KM, Olivares-Navarrete R, Wnt signaling modulates macrophage polarization and is regulated by biomaterial surface properties, *Biomaterials* 243 (2020). 10.1016/j.biomaterials.2020.119920.
- [23]. Avery D, Morandini L, Celt N, Bergery L, Simmons J, Martin RK, Donahue HJ, Olivares-Navarrete R, Immune cell response to orthopedic and craniofacial biomaterials depends on biomaterial composition, *Acta Biomater* 161 (2023) 285–297. 10.1016/J.ACTBIO.2023.03.007. [PubMed: 36905954]

- [24]. Jhunjhunwala S, Neutrophils at the Biological-Material Interface, *ACS Biomater Sci Eng* 4 (2018) 1128–1136. 10.1021/ACSBOMATERIALS.6B00743/ASSET/IMAGES/LARGE/AB-2016-00743J\_0003.JPEG. [PubMed: 33418651]
- [25]. Hotchkiss KM, Clark NM, Olivares-Navarrete R, Macrophage response to hydrophilic biomaterials regulates MSC recruitment and T-helper cell populations, *Biomaterials* 182 (2018) 202–215. 10.1016/J.BIOMATERIALS.2018.08.029. [PubMed: 30138783]
- [26]. Abaricia JO, Farzad N, Heath TJ, Simmons J, Morandini L, Olivares-Navarrete R, Control of innate immune response by biomaterial surface topography, energy, and stiffness, *Acta Biomater* 133 (2021) 58–73. 10.1016/J.ACTBIO.2021.04.021. [PubMed: 33882355]
- [27]. Avery D, Morandini L, Gabric M, Sheakley L, Peralta M, Donahue HJ, Martin RK, Olivares-Navarrete R, Contribution of  $\alpha\beta$  T cells to macrophage polarization and MSC recruitment and proliferation on titanium implants, *Acta Biomater* 169 (2023) 605–624. 10.1016/J.ACTBIO.2023.07.052. [PubMed: 37532133]
- [28]. Flaherty S, Reynolds JM, Mouse Naïve CD4+ T Cell Isolation and In vitro Differentiation into T Cell Subsets, *J Vis Exp* 2015 (2015) 52739. 10.3791/52739.
- [29]. King S, Klineberg I, Brennan-Speranza TC, Adipose Tissue Dysfunction: Impact on Bone and Osseointegration, *Calcif Tissue Int* 110 (2022) 32–40. 10.1007/S00223-021-00899-0/TABLES/1. [PubMed: 34374815]
- [30]. Silva BLG, Sánchez-Puetate JC, Pinotti FE, Marcantonio CC, Pedroso GG, Junior EM, Marcantonio RAC, Influence of obesity on osseointegration of implants with different surface treatments: A preclinical study, *Clin Implant Dent Relat Res* 25 (2023) 919–928. 10.1111/CID.13234. [PubMed: 37309706]
- [31]. Avery D, Morandini L, Sheakley LS, Shah AH, Bui L, Abaricia JO, Olivares-Navarrete R, Canonical Wnt signaling enhances pro-inflammatory response to titanium by macrophages, *Biomaterials* 289 (2022) 121797. 10.1016/J.BIOMATERIALS.2022.121797. [PubMed: 36156410]
- [32]. Morandini L, Avery D, Angeles B, Winston P, Martin RK, Donahue HJ, Olivares-Navarrete R, Reduction of neutrophil extracellular traps accelerates inflammatory resolution and increases bone formation on titanium implants, *Acta Biomater* (2023). 10.1016/J.ACTBIO.2023.05.016.
- [33]. Nguyen-Phuong T, Seo S, Cho BK, Lee JH, Jang J, Park CG, Determination of progressive stages of type 2 diabetes in a 45% high-fat diet-fed C57BL/6J mouse model is achieved by utilizing both fasting blood glucose levels and a 2-hour oral glucose tolerance test, *PLoS One* 18 (2023). 10.1371/JOURNAL.PONE.0293888.
- [34]. Hotchkiss KM, Reddy GB, Hyzy SL, Schwartz Z, Boyan BD, Olivares-Navarrete R, Titanium surface characteristics, including topography and wettability, alter macrophage activation, *Acta Biomater* 31 (2016) 425–434. 10.1016/J.ACTBIO.2015.12.003. [PubMed: 26675126]
- [35]. Abaricia JO, Shah AH, Musselman RM, Olivares-Navarrete R, Hydrophilic titanium surfaces reduce neutrophil inflammatory response and NETosis, *Biomater Sci* 8 (2020) 2289–2299. 10.1039/C9BM01474H. [PubMed: 32163073]
- [36]. Heinecke JW, Goldberg IJ, Myeloperoxidase: A Therapeutic Target for Preventing Insulin Resistance and the Metabolic Sequelae of Obesity?, *Diabetes* 63 (2014) 4001–4003. 10.2337/DB14-1273. [PubMed: 25414015]
- [37]. Mincham KT, Panchal K, Hart PH, Lucas RM, Feelisch M, Weller RB, Matthews VB, Strickland DH, Gorman S, Metabolic dysfunction induced by a high-fat diet modulates hematopoietic stem and myeloid progenitor cells in brown adipose tissue of mice, *Immunol Cell Biol* 99 (2021) 749–766. 10.1111/IMCB.12460. [PubMed: 33866598]
- [38]. Klebanoff SJ, Oxygen metabolism and the toxic properties of phagocytes, *Ann Intern Med* 93 (1980) 480–489. 10.7326/0003-4819-93-3-480. [PubMed: 6254418]
- [39]. Huang JY, Zhou QL, Huang CH, Song Y, Sharma AG, Liao Z, Zhu K, Massidda MW, Jamieson RR, Zhang JY, Tenen DG, Jiang ZY, Neutrophil elastase regulates emergency myelopoiesis preceding systemic inflammation in diet-induced obesity, *Journal of Biological Chemistry* 292 (2017) 4770–4776. 10.1074/jbc.C116.758748. [PubMed: 28202548]
- [40]. Aratani Y, Myeloperoxidase: Its role for host defense, inflammation, and neutrophil function, *Arch Biochem Biophys* 640 (2018) 47–52. 10.1016/J.ABB.2018.01.004. [PubMed: 29336940]

- [41]. Wu D, Ren Z, Pae M, Han SN, Meydani SN, Diet-induced obesity has a differential effect on adipose tissue and macrophage inflammatory responses of young and old mice, *BioFactors* 39 (2013) 326–333. 10.1002/BIOF.1075. [PubMed: 23345024]
- [42]. Kim MS, Choi MS, Han SN, High fat diet-induced obesity leads to proinflammatory response associated with higher expression of NOD2 protein, *Nutr Res Pract* 5 (2011) 219–223. 10.4162/NRP.2011.5.3.219. [PubMed: 21779525]
- [43]. Timpson NJ, Nordestgaard BG, Harbord RM, Zacho J, Frayling TM, Tybjaerg-Hansen A, Smith GD, C-reactive protein levels and body mass index: elucidating direction of causation through reciprocal Mendelian randomization, *Int J Obes (Lond)* 35 (2011) 300–308. 10.1038/IJO.2010.137. [PubMed: 20714329]
- [44]. Li Q, Wang Q, Xu W, Ma Y, Wang Q, Eatman D, You S, Zou J, Champion J, Zhao L, Cui Y, Li W, Deng Y, Ma L, Wu B, Wang G, Zhang X, Wang Q, Bayorh MA, Song Q, C-Reactive Protein Causes Adult-Onset Obesity Through Chronic Inflammatory Mechanism, *Front Cell Dev Biol* 8 (2020). 10.3389/FCELL.2020.00018.
- [45]. Devaraj S, Jialal I, C-reactive protein polarizes human macrophages to an M1 Phenotype and inhibits transformation to the M2 phenotype, *Arterioscler Thromb Vasc Biol* 31 (2011) 1397–1402. 10.1161/ATVBAHA.111.225508/FORMAT/EPUB. [PubMed: 21415385]
- [46]. Lumeng CN, Maillard I, Saltiel AR, T-ing up inflammation in fat, *Nature Medicine* 2009 15:8 15 (2009) 846–847. 10.1038/nm0809-846.
- [47]. Harford KA, Reynolds CM, McGillicuddy FC, Roche HM, Fats, inflammation and insulin resistance: insights to the role of macrophage and T-cell accumulation in adipose tissue, *Proceedings of the Nutrition Society* 70 (2011) 408–417. 10.1017/S0029665111000565. [PubMed: 21835098]
- [48]. Reinke S, Geissler S, Taylor WR, Schmidt-Bleek K, Juelke K, Schwachmeyer V, Dahne M, Hartwig T, Akyüz L, Meisel C, Unterwalder N, Singh NB, Reinke P, Haas NP, Volk HD, Duda GN, Terminally differentiated CD8<sup>+</sup> T cells negatively affect bone regeneration in humans, *Sci Transl Med* 5 (2013). 10.1126/SCITRANSLMED.3004754.
- [49]. Richard C, Wadowski M, Goruk S, Cameron L, Sharma AM, Field CJ, Individuals with obesity and type 2 diabetes have additional immune dysfunction compared with obese individuals who are metabolically healthy, *BMJ Open Diabetes Res Care* 5 (2017) e000379. 10.1136/BMJDR-2016-000379.
- [50]. Vargas R, Ryder E, Diez-Ewald M, Mosquera J, Durán A, Valero N, Pedrañez A, Peña C, Fernández E, Increased C-reactive protein and decreased Interleukin-2 content in serum from obese individuals with or without insulin resistance: Associations with leukocyte count and insulin and adiponectin content, *Diabetes & Metabolic Syndrome: Clinical Research & Reviews* 10 (2016) S34–S41. 10.1016/J.DSX.2015.09.007.
- [51]. Sánchez-Margalet V, Fernández-Riejos P, Najib S, Santos-Alvarez J, Martín-Romero C, Pérez-Pérez A, González-Yanes C, Role of Leptin in the Activation of Immune Cells, *Mediators Inflamm* 2010 (2010) 568343. 10.1155/2010/568343. [PubMed: 20368778]
- [52]. Kiernan K, MacIver NJ, The Role of the Adipokine Leptin in Immune Cell Function in Health and Disease, *Front Immunol* 11 (2021) 622468. 10.3389/FIMMU.2020.622468/BIBTEX. [PubMed: 33584724]
- [53]. Hong SK, Choo EH, Ihm SH, Chang KY, High fat diet downregulates regulatory T cells in the myocardium of spontaneous hypertensive rats, *Nutrition, Metabolism and Cardiovascular Diseases* 29 (2019) 1254–1260. 10.1016/J.NUMECD.2019.08.004.
- [54]. Kucharska AM, Pyrzak B, Demkow U, Regulatory T Cells in Obesity, *Adv Exp Med Biol* 866 (2015) 35–40. 10.1007/5584\_2015\_147. [PubMed: 26022902]
- [55]. Matarese G, Procaccini C, De Rosa V, Horvath TL, La Cava A, Regulatory T cells in obesity: the leptin connection, *Trends Mol Med* 16 (2010) 247–256. 10.1016/J.MOLMED.2010.04.002. [PubMed: 20493774]
- [56]. Tencerova M, Figeac F, Ditzel N, Taipaleenmäki H, Nielsen TK, Kassem M, High-Fat Diet-Induced Obesity Promotes Expansion of Bone Marrow Adipose Tissue and Impairs Skeletal Stem Cell Functions in Mice, *Journal of Bone and Mineral Research* 33 (2018) 1154–1165. 10.1002/JBMR.3408. [PubMed: 29444341]

- [57]. Kwon EY, Shin SK, Cho YY, Jung UJ, Kim E, Park T, Park JHY, Yun JW, McGregor RA, Park YB, Choi MS, Time-course microarrays reveal early activation of the immune transcriptome and adipokine dysregulation leads to fibrosis in visceral adipose depots during diet-induced obesity, *BMC Genomics* 13 (2012) 1–16. 10.1186/1471-2164-13-450/FIGURES/7. [PubMed: 22214261]
- [58]. Ambrosi TH, Scialdone A, Graja A, Gohlke S, Jank AM, Bocian C, Woelk L, Fan H, Logan DW, Schürmann A, Saraiva LR, Schulz TJ, Adipocyte Accumulation in the Bone Marrow during Obesity and Aging Impairs Stem Cell-Based Hematopoietic and Bone Regeneration, *Cell Stem Cell* 20 (2017) 771. 10.1016/J.STEM.2017.02.009. [PubMed: 28330582]
- [59]. De Assis-Ferreira A, Saldanha-Gama R, De Brito NM, Renovato-Martins M, Simões RL, Barja-Fidalgo C, Da Silva SV, Obesity enhances the recruitment of mesenchymal stem cells to visceral adipose tissue, *J Mol Endocrinol* 67 (2021) 15–26. 10.1530/JME-20-0229. [PubMed: 34045365]
- [60]. Cai F, Yusufu A, Liu K, Chen W, Zhao R, Liu Y, Liu Y, High-fat diet causes undesirable bone regeneration by altering the bone marrow environment in rats, *Front Endocrinol (Lausanne)* 14 (2023) 1088508. 10.3389/FENDO.2023.1088508/BIBTEX. [PubMed: 37056669]
- [61]. da Silva SV, Renovato-Martins M, Ribeiro-Pereira C, Citelli M, Barja-Fidalgo C, Obesity modifies bone marrow microenvironment and directs bone marrow mesenchymal cells to adipogenesis, *Obesity* 24 (2016) 2522–2532. 10.1002/OBY.21660. [PubMed: 27753270]
- [62]. Yu L, Tu Q, Han Q, Zhang L, Sui L, Zheng L, Meng S, Tang Y, Xuan D, Zhang J, Murray D, Shen Q, Cheng J, Kim SH, Dong LQ, Valverde P, Cao X, Chen J, Adiponectin Regulates Bone Marrow Mesenchymal Stem Cell Niche Through a Unique Signal Transduction Pathway: An Approach for Treating Bone Disease in Diabetes, *Stem Cells* 33 (2015) 240–252. 10.1002/STEM.1844. [PubMed: 25187480]
- [63]. Wu X, Zhu D, Shi L, Tu Q, Yu Y, Chen J, AdipoRon accelerates bone repair of calvarial defect in diet-induced obesity mice, *Heliyon* 9 (2023). 10.1016/J.HELİYON.2023.E13975.
- [64]. Gao F, Lv TR, Zhou JC, Qin XD, Effects of obesity on the healing of bone fracture in mice, *J Orthop Surg Res* 13 (2018). 10.1186/S13018-018-0837-7.
- [65]. Halade GV, El Jamali A, Williams PJ, Fajardo RJ, Fernandes G, Obesity-mediated inflammatory microenvironment stimulates osteoclastogenesis and bone loss in mice, *Exp Gerontol* 46 (2011) 43–52. 10.1016/J.EXGER.2010.09.014. [PubMed: 20923699]
- [66]. Watanabe K, Maki K, Serum leptin levels negatively correlate with trabecular bone mineral density in high-fat diet-induced obesity mice, *Article in Journal of Musculoskeletal & Neuronal Interactions* (2012). <https://www.researchgate.net/publication/225080436> (accessed October 8, 2024).
- [67]. Cao JJ, Sun L, Gao H, Diet-induced obesity alters bone remodeling leading to decreased femoral trabecular bone mass in mice, *Ann N Y Acad Sci* 1192 (2010) 292–297. 10.1111/J.1749-6632.2009.05252.X. [PubMed: 20392249]
- [68]. Hata M, Andriessen EMMA, Hata M, Diaz-Marin R, Fournier F, Crespo-Garcia S, Blot G, Juneau R, Pilon F, Dejda A, Guber V, Heckel E, Daneault C, Calderon V, Des Rosiers C, Melichar HJ, Langmann T, Joyal JS, Wilson AM, Sapieha P, Past history of obesity triggers persistent epigenetic changes in innate immunity and exacerbates neuroinflammation, *Science* 379 (2023) 45–62. 10.1126/SCIENCE.ABJ8894. [PubMed: 36603072]
- [69]. Wang X, Zhu H, Snieder H, Su S, Munn D, Harshfield G, Maria BL, Dong Y, Treiber F, Gutin B, Shi H, Obesity related methylation changes in DNA of peripheral blood leukocytes, *BMC Med* 8 (2010) 1–8. 10.1186/1741-7015-8-87/TABLES/5. [PubMed: 20051100]
- [70]. Raghuraman S, Donkin I, Versteyhe S, Barrès R, Simar D, The Emerging Role of Epigenetics in Inflammation and Immunometabolism, *Trends in Endocrinology and Metabolism* 27 (2016) 782–795. 10.1016/j.tem.2016.06.008. [PubMed: 27444065]
- [71]. Amarasekera M, Prescott SL, Palmer DJ, Nutrition in early life, immune-programming and allergies: the role of epigenetics, (n.d.).
- [72]. Soriano-Baguet L, Brenner D, Metabolism and epigenetics at the heart of T cell function, *Trends Immunol* 44 (2023) 231–244. 10.1016/j.it.2023.01.002. [PubMed: 36774330]
- [73]. Zatterale F, Raciti GA, Prevezano I, Leone A, Campitelli M, De Rosa V, Beguinot F, Parrillo L, Epigenetic Reprogramming of the Inflammatory Response in Obesity and Type 2 Diabetes, *Biomolecules* 2022, Vol. 12, Page 982 12 (2022) 982. 10.3390/BIOM12070982.

- [74]. Wang X, Cao Q, Yu L, Shi H, Xue B, Shi H, Epigenetic regulation of macrophage polarization and inflammation by DNA methylation in obesity, JCI Insight 1 (2016) 87748. 10.1172/JCI.INSIGHT.87748. [PubMed: 27882346]

Author Manuscript

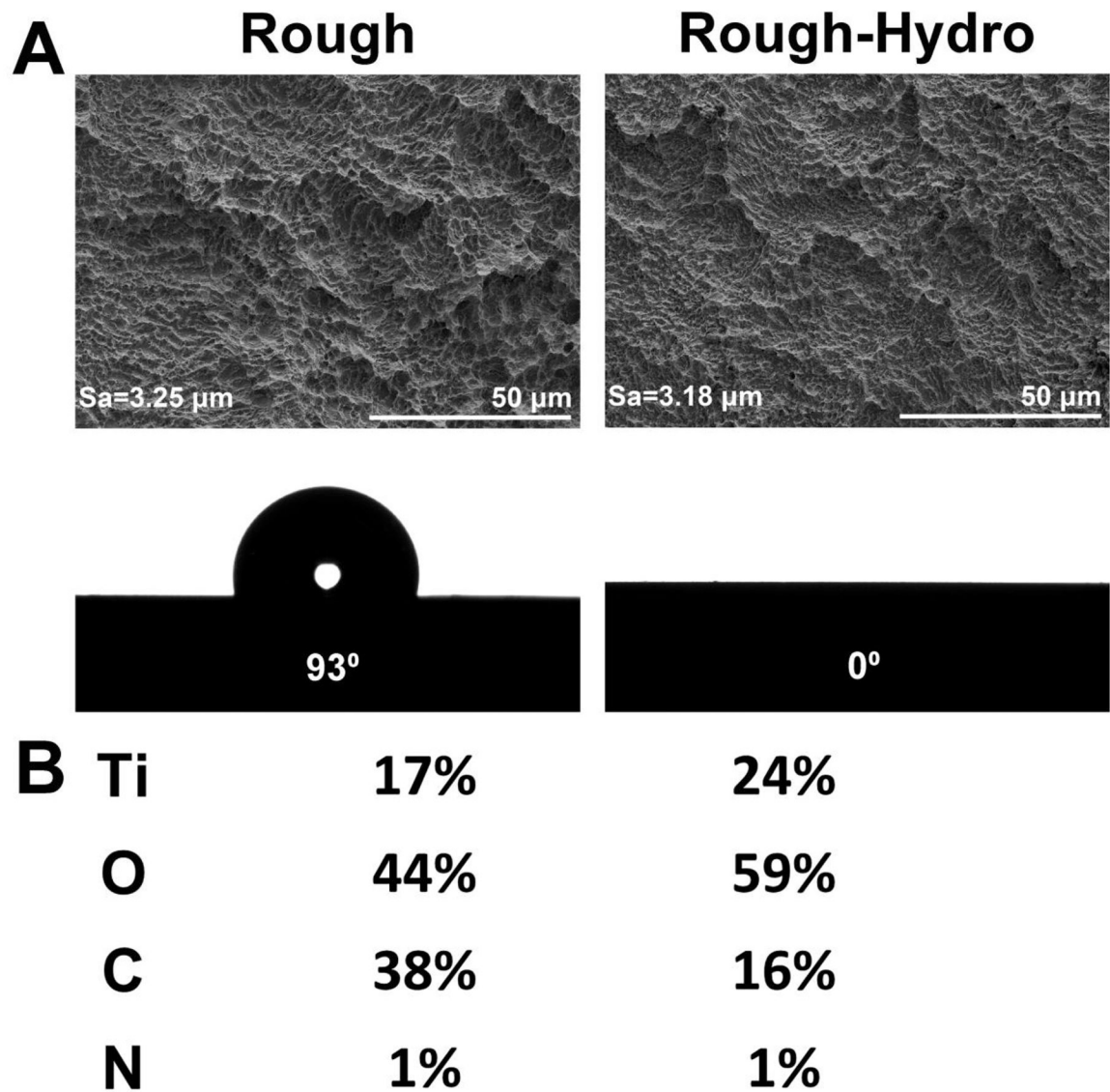
Author Manuscript

Author Manuscript

Author Manuscript

**Statement of Significance**

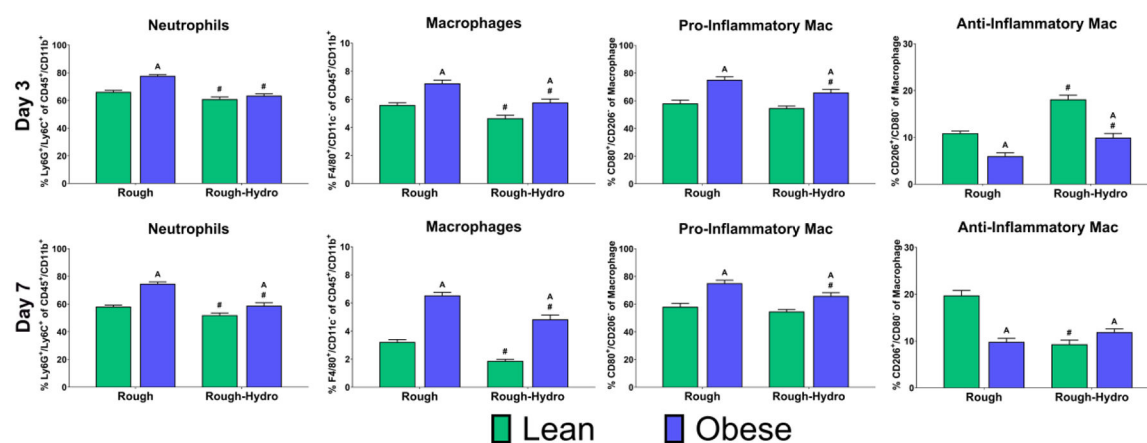
Obesity has been shown to significantly alter physiological processes, including the behavior of immune cells, inducing a state of systemic chronic inflammation. Our study demonstrates that obesity-induced via a high-fat diet alters immune cell response to implanted biomaterials, with increased pro-inflammatory response and attenuated immunomodulation resulting in decreased biomaterial integration.



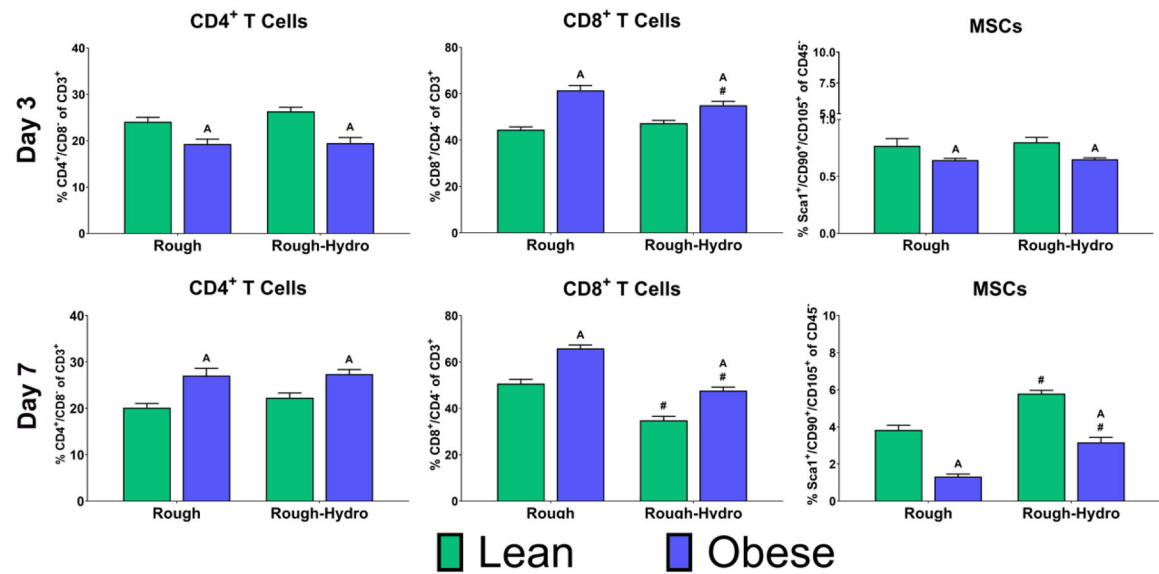
**Figure 1:**

Characterization of rough and rough hydrophilic Ti samples. (A) Surface topography analyzed qualitatively via scanning electron microscopy B) Measurement of water contact angle (C) Chemical composition of the material oxide layer for each sample.

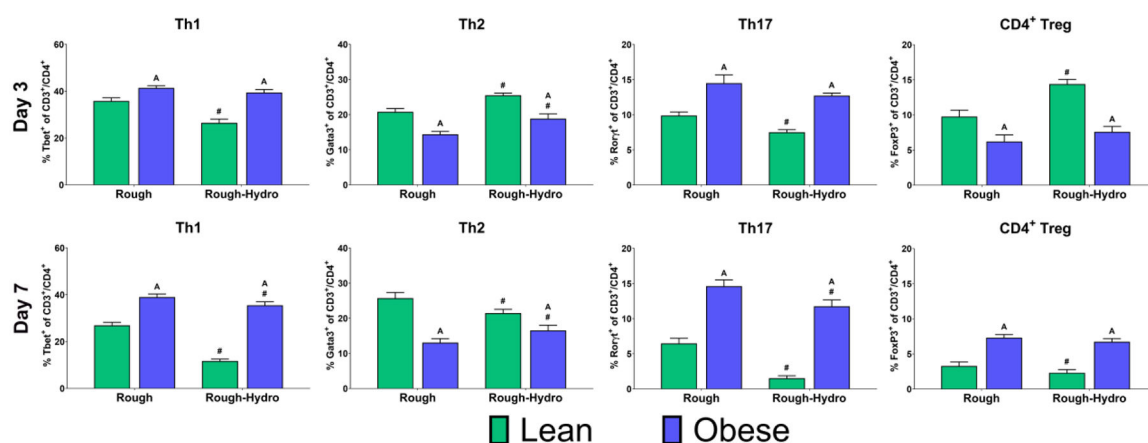


**Figure 2:**

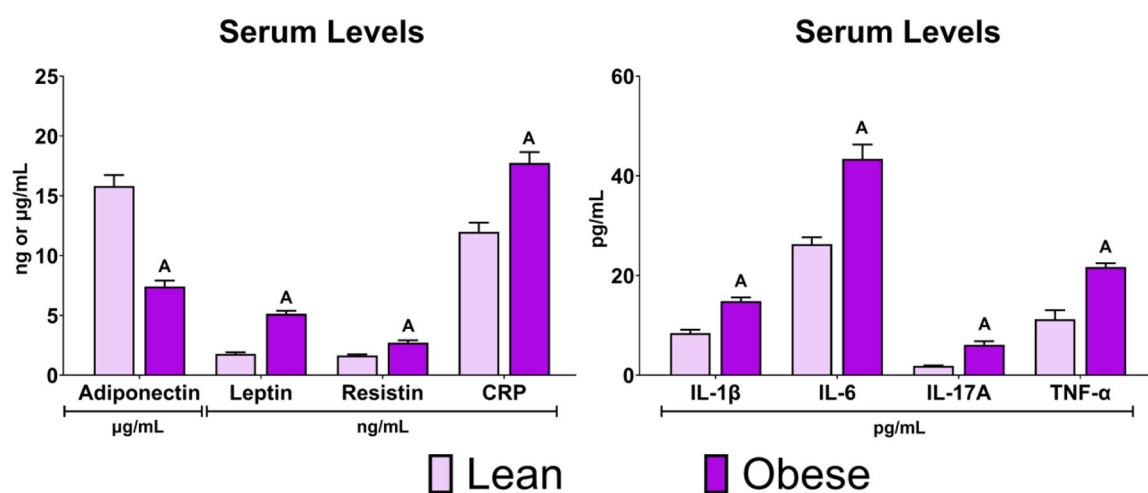
Obese mice have significantly different populations of neutrophils, total macrophages, pro-inflammatory macrophages, and anti-inflammatory macrophages in response to modified Ti implants. Flow cytometry analysis of peri-implant bone marrow at 3- and 7-days post-implantation in obese and lean mice.  $p < 0.05$ : A vs. lean, # vs. rough Ti.

**Figure 3:**

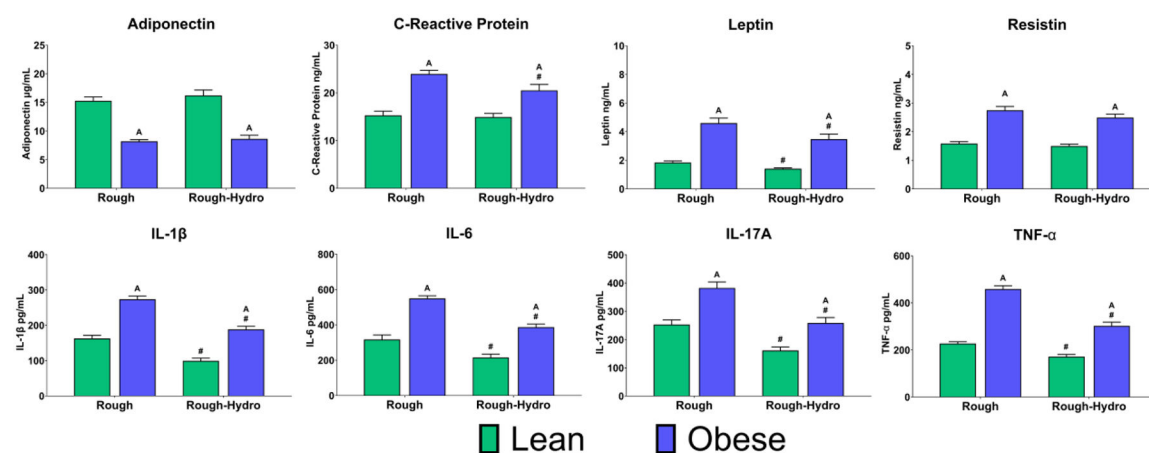
Obese mice have altered recruitment of CD4<sup>+</sup> T cells, CD8<sup>+</sup> T cells, and MSCs in response to modified Ti implants. Flow cytometry analysis of peri-implant bone marrow at 3- and 7-days post-implantation in obese and lean mice.  $p < 0.05$ : A vs. lean, # vs. rough Ti.

**Figure 4:**

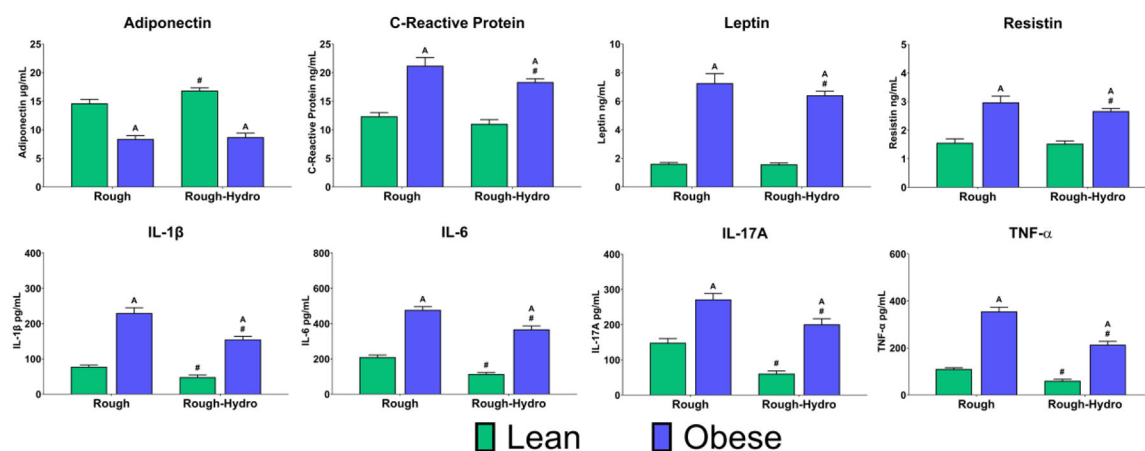
CD4<sup>+</sup> T cell polarization is impacted by obesity, skewing polarization towards Th1/Th17 phenotypes in response to modified Ti implants. Flow cytometry analysis of peri-implant bone marrow at 3- and 7-days post-implantation in obese and lean mice.  $p < 0.05$ : A vs. lean, # vs. rough Ti.



**Figure 5:**  
Baseline levels of serum adipokines and pro-inflammatory cytokines are altered in obese mice prior to implantation. Cytokine analysis of serum from obese and lean mice.  $p < 0.05$ : A vs. lean.

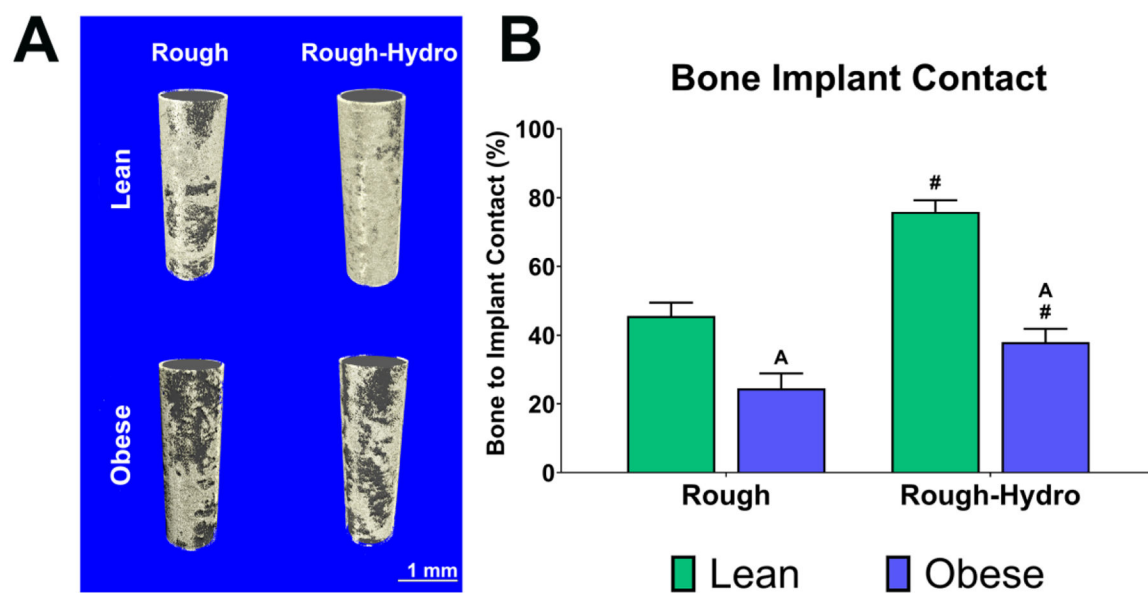
**Figure 6:**

Serum levels of adipokines and pro-inflammatory cytokines are significantly impacted in obese mice in response to modified Ti implants. Cytokine analysis of serum from obese and lean mice at 3-days post-implantation  $p < 0.05$ : A vs. lean, # vs. rough Ti.



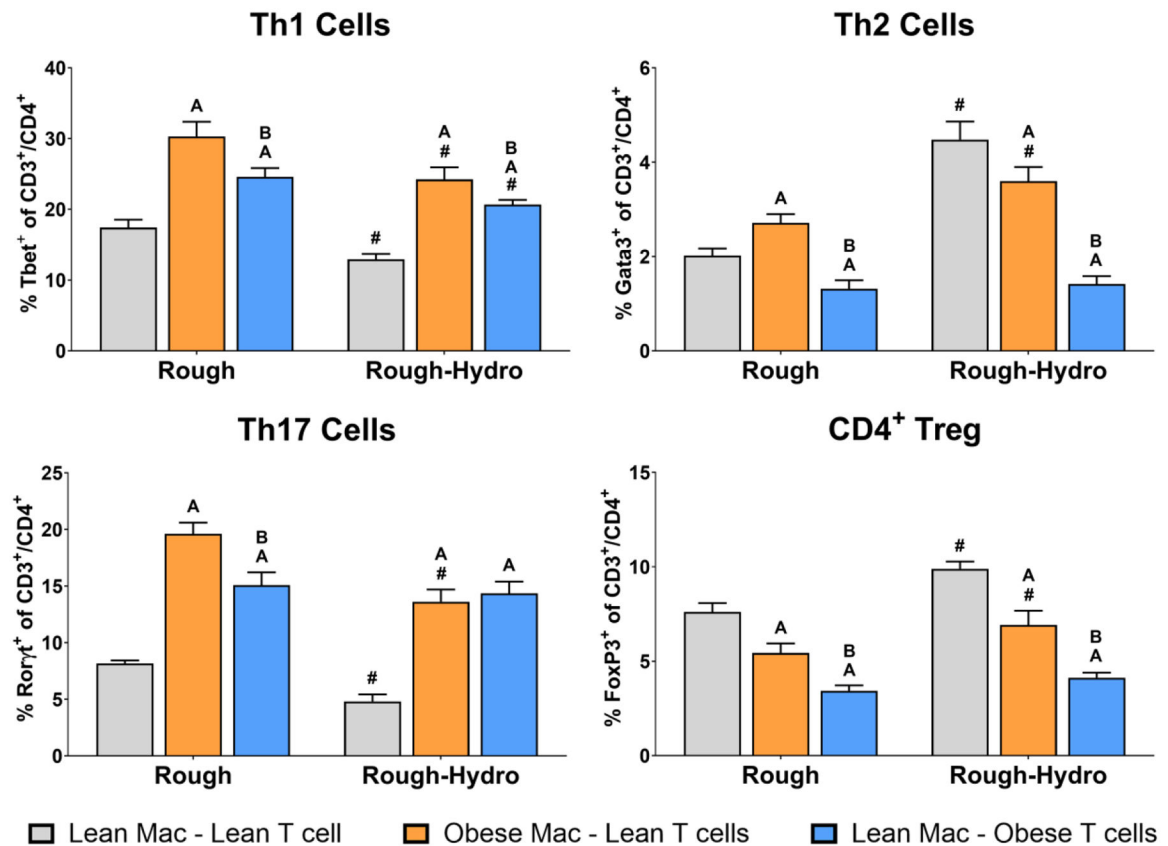
**Figure 7:**

Secretion of adipokines and pro-inflammatory cytokines in serum are significantly altered in obese mice in response to modified Ti implants. Cytokine analysis of serum from obese and lean mice at 7-days post-implantation.  $p < 0.05$ : A vs. lean, # vs. rough Ti.

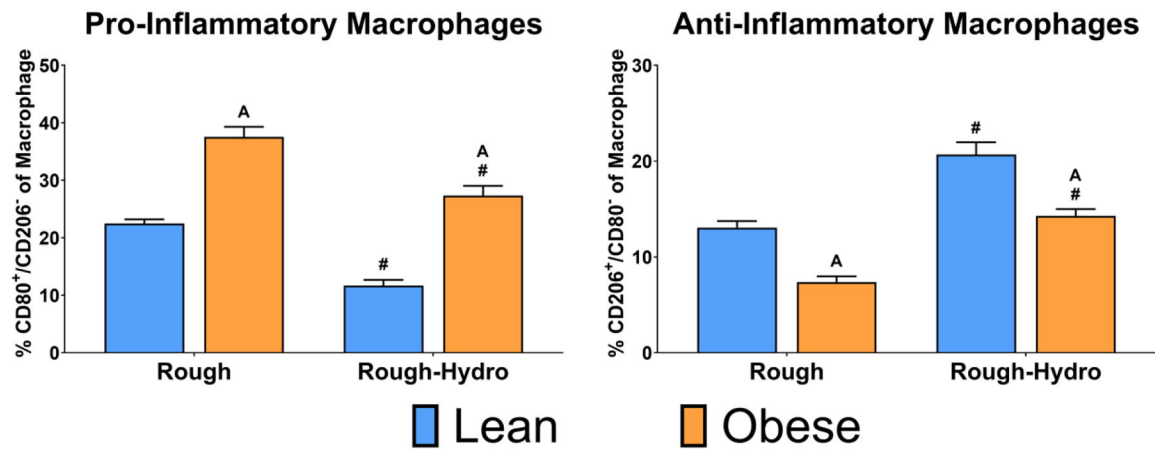


**Figure 8:** Bone implant contact is significantly lower in obese mice. Analysis of bone-implant contact around modified Ti implants in obese and lean mice at 21-days post-implantation.  $p < 0.05$ : A vs. lean, # vs. rough Ti.



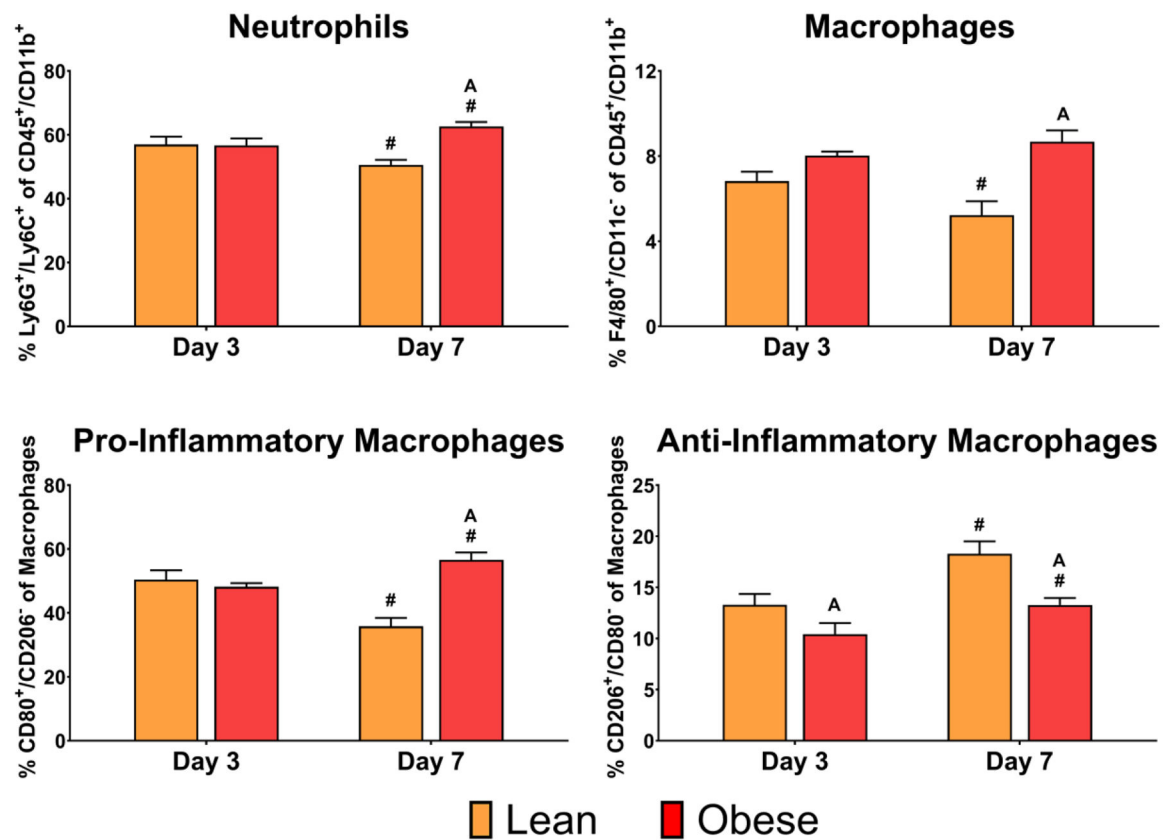
**Figure 9:**

CD4<sup>+</sup> T cell subset polarization is skewed by obesity *in vitro*. Flow cytometry analysis of CD4<sup>+</sup> T cell subsets from macrophage-T cell cocultures from obese or lean mice on modified Ti surfaces. p<0.05: A vs. Lean Mac-Lean T Cells, B vs. Obese Mac-Lean T Cells, # vs. rough Ti.



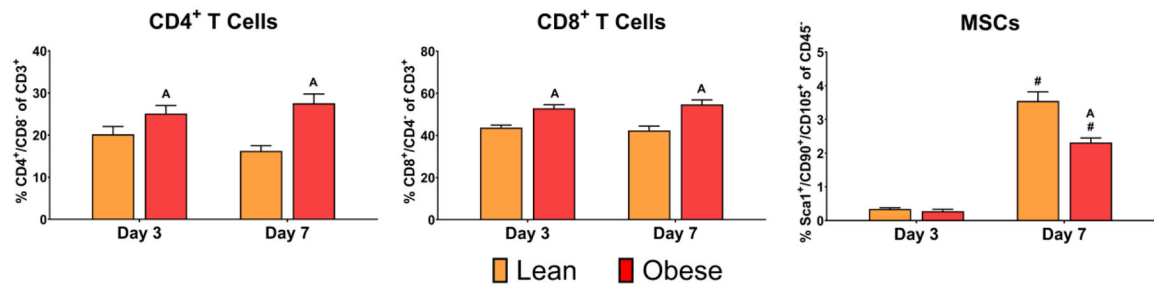
**Figure 10:**

Macrophage polarization is altered by obesity *in vitro*. Flow cytometry analysis of pro- and anti-inflammatory macrophages from either obese or lean mice cultured on modified Ti surfaces.  $p < 0.05$ : A vs. lean, # vs. rough Ti.



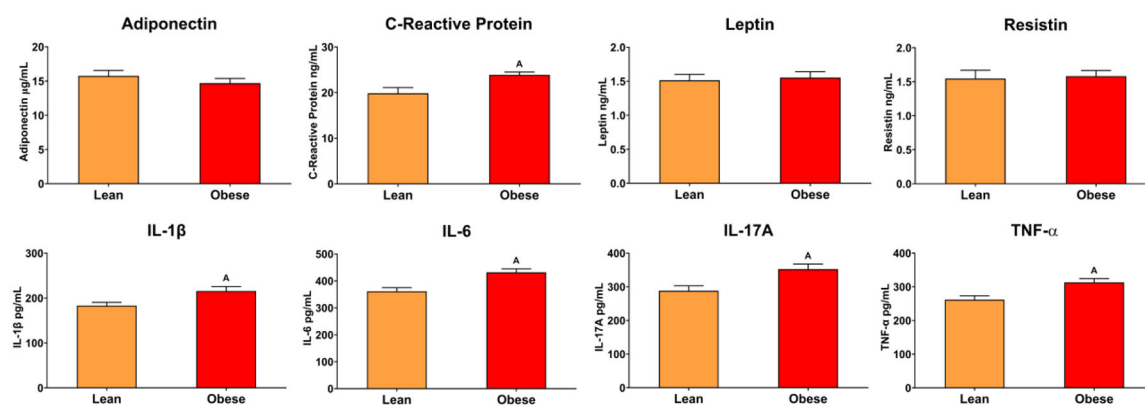
**Figure 11:**

Adoptive transfer of bone marrow from obese mice significantly skews populations of neutrophils, total macrophages, pro-inflammatory macrophages, and anti-inflammatory macrophages in response to modified Ti implants. Flow cytometry analysis of peri-implant bone marrow at 3- and 7-days post-implantation in mice receiving bone marrow cells from either obese or lean mice.  $p < 0.05$ : A vs. lean, # vs. day 3.

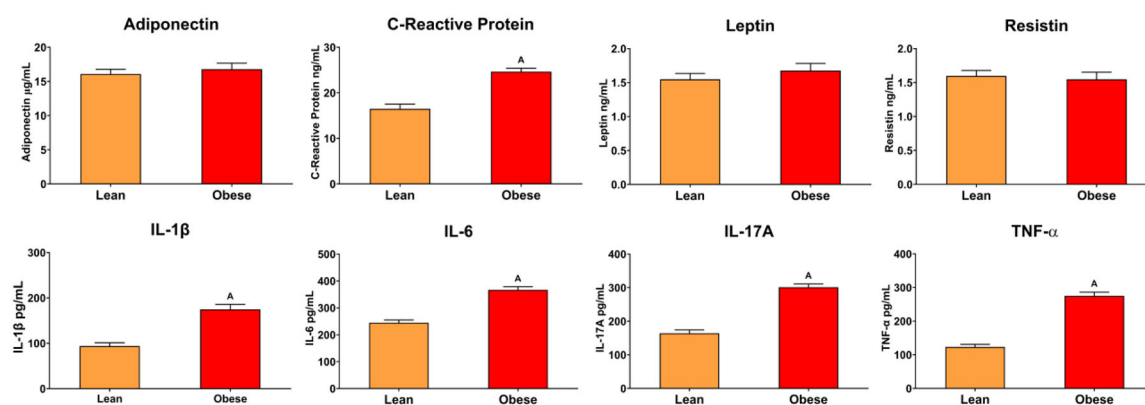


**Figure 12:**

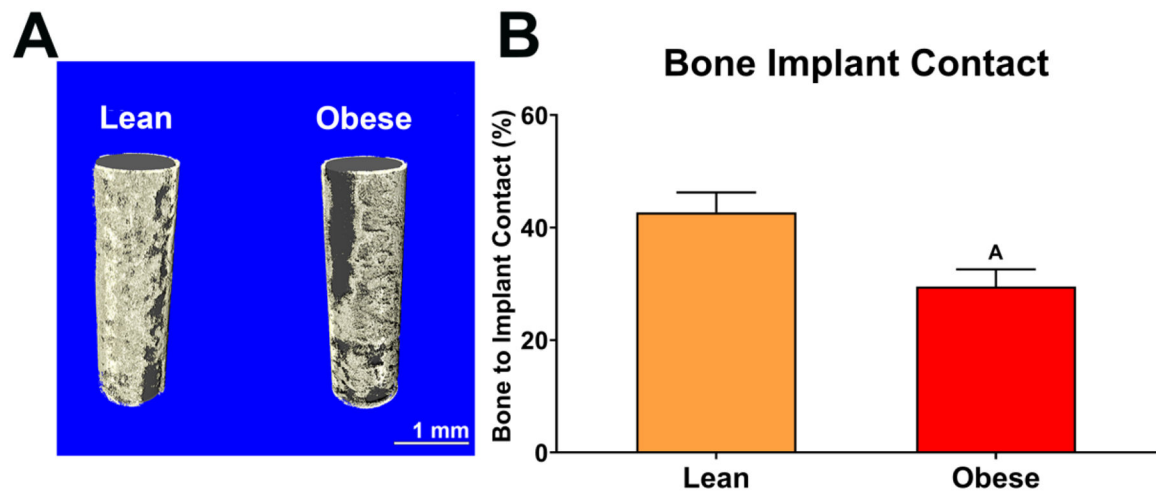
Adoptive transfer of bone marrow from obese mice significantly alters recruitment of CD4<sup>+</sup> T cells, CD8<sup>+</sup> T cells, and MSCs in response to modified Ti implants. Flow cytometry analysis of peri-implant bone marrow at 3- and 7-days post-implantation in mice receiving bone marrow cells from either obese or lean mice.  $p < 0.05$ : A vs. lean, # vs. day 3.



**Figure 13:** Serum levels of adipokines and pro-inflammatory cytokines are impacted by adoptive transfer of bone marrow cells from obese mice. Cytokine analysis of serum in mice receiving bone marrow cells from either obese or lean mice at 3-days post-implantation.  $p < 0.05$ : A vs. lean.



**Figure 14:** Secretion of adipokines and pro-inflammatory cytokines in serum are significantly impacted by adoptive transfer of bone marrow cells from obese mice. Cytokine analysis of serum in mice receiving bone marrow cells from either obese or lean mice at 7-days post-implantation. p<0.05: A vs. lean.



**Figure 15:**

Bone implant contact is significantly lower in mice adoptively transferred bone marrow cells from obese mice. Analysis of bone-implant contact around modified Ti implants mice receiving bone marrow cells from either obese or lean mice at 21-days post-implantation.  $p < 0.05$ : A vs. lean.